UNIVERSITI TUNKU ABDUL RAHMAN
LEE KONG CHIAN FACULTY OF ENGINEERING AND SCIENCE
LABORATORY SAFETY RULES

The following rules must be obeyed by all students in the Computer laboratories of the faculty. Willful or repeated inadvertent noncompliance may result in dismissal or suspension from the laboratories.

I. No entry without permission:

i) Outsiders are not allowed to enter the laboratory without permission.

ii) No student is allowed to enter the laboratory unless permission has been given by a laboratory staff or a lecturer.

II. At work in the laboratory:

i) The computer facilities at the laboratory are to be used for learning purposes only.

ii) Wear your I.D. cards at all times in a proper visible position whenever you come into the computer laboratory.

iii) It is compulsory to follow dress code of UTAR
   a) Covered shoes (no slippers or flip flops)
   b) Long pants and tidy attire

iv) Do not eat or drink in the laboratory.

v) Do not tamper with any computers, printers, or any other associated equipment. If there are problems or faults, report immediately to the staff at the laboratory.

vi) Do not remove any equipment or document from the computer laboratories.

vii) Do not alter the default hardware and software setup of the computers and printers in the computer laboratories.

viii) Do not copy, rename, change, examine or delete files or information belonging to any other user of the University.

ix) Do not install any unauthorized or unlicensed software into the PC of computer laboratories.

x) Do not make any noise, chit-chat, hold group discussions or disturb other users at the laboratory in any way.

xi) Silent your hand phone when you are in the computer laboratories.

xii) Any equipment broken or damaged must be reported immediately to the laboratory staff.

III. Before leaving the laboratory:

i) No student is allowed to take away any equipment, tools or other property of the laboratory without permission.

ii) Please check and remember to take all your personal belongings such as handphone, wallet, pendrive, hard disk, book, bottle, sweeter, and etc.

iii) Lab staffs are not responsible for lost of personal belongings.

Universiti Tunku Abdul Rahman

Faculty	**Lee Kong Chian Faculty of Engineering and Science**
Department:	**Department of Electrical and Electronics Engineering**
Unit Code and Name	**UEEN 1043 Object-Oriented Concepts and Programming Techniques**
Experiment No.:	**2 / 7**
Title of Experiment:	**Fundamentals of A Program**
Laboratory Room No. and Name:	**Computer lab**
Experiment Duration (hour):	**3 hours**
Number of Student per Group	**1 student**

Objectives

- To understand how to use the if statement
- To create loop constructs
- To define and apply functions

Equipment and Materials

Item Description	***Item category**	Quantity estimation (e.g. per set/group of student)
Personal computer	E	1 per student
Python interpreter (Anaconda Spyder)	S	1 per student

***Item category**	
SP	Sample or specimen
C	Consumable
CH	Chemical
W	Labware, glassware, tool, and components
E	Equipment
S	Software

Caution Statement (Optional)

FACULTY OF ENGINEERING AND SCIENCE
UNIVERSITI TUNKU ABDUL RAHMAN

UEEN 1043 Object-Oriented Concepts and Programming Techniques

Practical 2: Python's Program Constructs

Objectives

- To understand how to use the if statement
- To create loop constructs
- To define and apply functions

Equipment

- Personal computer with a Python interpreter (e.g. Anaconda Spyder)

Introduction

Launch Anaconda Spyder IDE for Python and write codes for the following exercises.

Exercises

2-1 If statement

Try out the following codes:

```
# Determine if a given number is odd or even
num = input("Enter a number: ")
mod = num % 2
if mod > 0:
    print("This is an odd number.")
else:
    print("This is an even number.")
```

You may also check for the occurrence of a substring in a long string.

```
if 'c' in 'abcde':
    print('Found c')
elif 'C' in 'abcde':
    print('Found C')
else:
    print('Cannot find C or c')
```

Or element in list.

```
if 2 in [1,2,3]:
     print('Yes')
```

2-2 For loops using range

```
for x in range(10):
    print(x)
for x in range(1,11):
    print(x)
for x in range(1,11,2): #skip by 2
    print(x)
for x in range(10,0,-1): #move in reverse
    print(x)

#Nested loops
for x in range(3):
    for y in range(3):
        print(x,y)
```

2-3 The Fibonacci Sequence

Fibonacci Sequence is defined as F(n) = F(n-1) + F(n-2), where F(0) = F(1) = 1. One way to generate the numbers as a series of a certain list size n.

```
a, b = 0, 1
while a < n:
    print(a)
    a, b = b, a+b
```

Study carefully and understand the order of execution in the above program. Compare the above solutions with the version from your lecture notes.

2-4 Function

A basic Python function has the following structure; to execute the function, just call it by passing in the right argument(s) and receiving the returned value(s).

```
def f(a, b):
    #body of function
    return a/2, 2*b

x, y = f(1, 2) #Call a function and receive its return values
```

Now you are asked to implement a function called isDivisor(x) which can satisfy the need of the program below:

```
num = int(input("Enter a number: "))
for x in range(1,num+1): # Find all divisors of a given number
```

```
    if isDivisor(x) :
        print(x)
```

2-5 More on list

Type the following codes and check their effects.

```
# Create/append/size/concatenate
array = []
array.append(1)
array.append('2')
array = array + [3]
array += ['4']
len(array)
vector=[5,6]
array + vector
2 in array
5 in array

# Index and slice
array[0]
array[:2]
array[2:]
array[1:2]

# List of lists
array = [[1, 2], [3, 4]]
array = []
array.append([])    #[[]]
array.append([])    #[[],[]]
array[0].append('a')#[['a'],[]]
array[0].append('b')#[['a', 'b'],[]]
array[1].append('c')#[['a', 'b'],['c']]
array[1].append('d')#[['a', 'b'],['c', 'd']]
array[1]            #['c', 'd']
array[1][0]         #'c'

# Converting string to list
str = '12345678'
list(str)
[list('12'), list('34')]

# Look up for all functionalities, e.g. remove/sort/count, of list by:
dir(list)
help(list)
```

2-6 Loop over lists

The for loop iterates over the items of a sequence (list or string) in the order they appear in the sequence.

```
words = ['Python', 'is', 'cool']
for e in words:
    print(e)
```

Another way to loop is to do list indexing.

```
words = ['Python', 'is', 'cool']
for e in range(len(words)):
    print(words[e])
```

List can be very useful for storing and grouping various kinds of data types together. So, an array of list is perfect for recording table values. We show below how to loop through list of lists.

```
data = [ ['A',1,2], ['B',3,4], ['C',5,6] ]
for a in data:
    for b in a:
        print(b)
```

Exercise

It is a must for you to demonstrate how to look for relevant knowledge/solution on the Internet on your own when you want to find answers beyond the lecture notes in order to complete the following.

A2

Write the algorithm solution in Python which is concise and general. The program should read the attached file called A2_input.csv and display the required ***analysis*** described in parts (a) to (e). These parts are ordered nicely to give you hints on how to solve the problem towards the final part. First, try to create a list of list structure to store the data from the csv file, see below.

```
[['Alan Tan', 'M', 39, 2000],
 ['Bradley', 'm', 34, 5448],
      :
 ['Zoe Yap', 'f', 25, 5769]]
```

Note that you must preprocess all the data to be clean, i.e. numbers should be numbers, and texts should be texts, in consistent casing, without redundant space or special characters like '\n', etc.

a) Print out all entries of the table as follow:

```
Alan Tan M 39 2000
      :
Zoe Yap F 25 5769
```

b) Print out all female names with their index numbers:

```
[2] Catherine
      :
```

```
[24] Zoe Yap
```

c) Find the richest man and woman in the list.

```
The richest man is …
The richest woman is …
```

d) Find the mean (i.e. average) of age, and mean of incomes in the list.

```
Mean age = …
Mean incomes = …
```

e) Find the median of the incomes in the list. Median is the item located in the middle of a sorted list. For even number of items, take the average of the two items closest to the virtual center of the sorted list. Print out the maximum and minimum of incomes as well.

```
Incomes:
     Median = …
     Max = …, Min = …
```

All real numbers should be displayed to a single decimal place. You can ***ONLY*** use Python keywords, ***for-loop***, built-in functions, and the methods of ***string*** and ***list*** in your solution. Do ***NOT import*** any library. Lastly, you are required to use ***function(s)*** to implement each part above, as well as any significant codes that repeat in structure for a modular programming design.